

Roshini Bollineni Ramesh

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Summary

Aspiring Software Engineer with hands-on experience in frontend and full-stack development. Proficient in building scalable web applications using React, Node.js, and MongoDB, with strong command over REST APIs, authentication, and deployment. Passionate about solving real-world problems and delivering user-focused solutions. Strong interest in scalable systems, data streaming, and impactful UI for real-time applications. Focused on optimizing ML models and minimizing API latency during deployment

Technical Skills

Programming Languages: Python, JavaScript, TypeScript, HTML, CSS, SQL, C, Java.

Libraries/Frameworks: React, Node.js, Express, Keras, TensorFlow, Scikit-learn, Axios.

Tools/Software: AWS, Salesforce, VS Code, GitHub, MongoDB, Postman, Jupyter, Android Studio, Power BI, Tableau, MS Office.

Soft Skills: Problem-Solving, Leadership, Presentation, Time Management, Adaptability, Event Planning.

Education

University of Massachusetts Lowell, MS in Computer Science

Sept 2023 – May 2025

Coursework: Artificial Intelligence, Algorithms, Data Mining, Web Systems, Deep Learning, Operating Systems, IoT, Computer Networks, Methods in Data Science.

CGPA: 3.7/4.0

Professional Experience

Data Science Intern, SmartKnower, India

May 2021 – June 2021

- Built an image classification model using TensorFlow achieving 95
- Conducted COVID-19 time-series analysis using Fbprophet with 85
- Visualized insights with Plotly; presented findings to stakeholders.
- **Technologies Used:** Python, Scikit-learn, Pandas, NumPy, Plotly, Fbprophet, TensorFlow.

Academic Experience

Student Support Assistant, UMass-Lowell, MA

Aug 2024 – May 2025

- Managed and resolved student support cases using Salesforce and Five9, ensuring timely academic assistance.
- Utilized Salesforce Lightning UI for interacting with cases, user records, and system workflows.
- Designed and maintained Salesforce dashboards and custom reports to monitor student success metrics.
- Created email templates and automated alerts using Salesforce tools, enhancing communication workflows.
- Worked with backend objects Cases, Leads, Users and supported data updates using data import tools.
- Collaborated with advisors and admin teams, demonstrating cross-functional coordination and problem-solving.

Orientation Leader, UMass-Lowell, MA

May 2024 – July 2024

- Organized and led orientation programs for incoming students, streamlining onboarding processes.
- Facilitated student engagement, resource familiarization, and registration assistance.
- Communicated regularly with stakeholders and addressed real-time issues with empathy and clarity.

Projects

Online Food Ordering Platform

- Developed a full-stack web app using React, Node.js, MongoDB, and Express.
- Implemented secure payment workflows using Stripe API.

- Designed and consumed RESTful APIs with Axios, secured app with JWT authentication.
- Configured automated setup scripts and deployed frontend/backend using scalable cloud services.
- Collaborated on API contracts and UI feedback cycles, improving UX and client-server integration.
- **Technologies Used:** React, TypeScript, Node.js, Express, MongoDB, Mongoose, Axios, JWT, Stripe, dotenv.

Adversarial Attacks and Defenses in Neural Networks

- Designed robust DNNs using Keras and TensorFlow; defended against PGD, BIM, C and W attacks.
- Implemented adversarial training, preprocessing, and masking techniques.
- Conducted extensive evaluation on MNIST and medical datasets.
- **Technologies Used:** Python, Keras, TensorFlow, NumPy, Pandas.

Cardiovascular Disease Prediction using Machine Learning

- Built and compared models (SVM, RF, Neural Networks, Naive Bayes) on clinical datasets.
- Preprocessed data with feature engineering and evaluated using precision, recall, and F1-score.
- Achieved high prediction accuracy using deep learning model.
- **Technologies Used:** Python, Keras, TensorFlow, Neural Networks, Scikit-learn, Pandas, NumPy.

Stress Detection using Empatica E4

- Processed biometric sensor data (EDA, HR, TEMP) from wearables.
- Trained LSTM model for real-time stress detection; achieved 92
- Employed data normalization, noise reduction, and live testing for real-world application.
- **Technologies Used:** Python, TensorFlow, Keras, NumPy, Pandas.

Intelligent Medical Box – IoT-Based Embedded System

- Built an Arduino-based medical reminder system using buzzer alerts to improve patient medication adherence.
- Programmed real-time buzzer control logic using Embedded C and Arduino IDE.
- Integrated a secure online portal to manage medication schedules and patient records.
- **Technologies Used:** Embedded C, Arduino IDE, Microcontrollers, IoT, Cloud Integration.